

Abstract Algebra

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Homework 3: Week 3 due Sep 14

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Herstein, section 2.4, page 64 : Questions 8,13,16,18,27,31,37

1. 8. If every right coset of H in G is a left coset of H in G , prove that $aHa^{-1} = H$ for all $a \in G$

Proof. We interpret the assumption as: For any right coset Ha in G , $Ha = aH$. Note that this also implies every left coset of H in G is a right coset. Now

$$\begin{aligned}x &\in aHa^{-1} \\&\Leftrightarrow x \in a(Ha^{-1}) \\&\Leftrightarrow x \in a(a^{-1}H) \\&\Leftrightarrow x \in (aa^{-1})H \\&\Leftrightarrow x \in H.\end{aligned}$$

□

2. 13. Find the orders of all the elements of U_{18} . Is U_{18} cyclic?

Proof. $U_{18} = \{[1], [5], [7], [11], [13], [17]\}$. We find order of each element:

$$\begin{aligned}\langle 1 \rangle &= \{[1]\} \\ \langle [5] \rangle &= \{[5], [7], [17], [13], [11], [1]\} \\ \langle [7] \rangle &= \{[7], [13], [1]\} \\ \langle [11] \rangle &= \{[11], [13], [17], [7], [5], [1]\} \\ \langle [13] \rangle &= \{[13], [7], [1]\} \\ \langle [17] \rangle &= \{[17], [1]\}.\end{aligned}$$

Hence, they have order 1, 6, 3, 6, 3, 2 respectively. The group U_{18} is hence cyclic since it is generated by $[5]$ and $[11]$. □

3. 16. If G is a finite abelian group and $a_1, \dots, a_n$ are all its elements, show that $x = a_1 a_2 \cdots a_n$ must satisfy $x^2 = e$.

Proof. Without loss of generality we assume $a_1, \dots, a_n$ are distinct. Let $a'_1 a'_2 \dots a'_n$ be a permutation of $a_1 a_2 \dots a_n$. Because G is finite and abelian,

permutation does not change the product. Hence $x^2 = (a_1 a_2 \dots a_n)(a_1 a_2 \dots a_n) = (a_1 a_2 \dots a_n)(a'_1 a'_2 \dots a'_n)$.

Now by the inverse axiom, there must exist some $a \in G$ such that $a_1 a = e$. Without loss of generality we can call it a'_1 . Hence

$$(a_1 a_2 \dots a_n)(a'_1 a'_2 \dots a'_n) = (a_1 a'_1)(a_2 \dots a_n)(a'_2 \dots a'_n) = (a_2 \dots a_n)(a'_2 \dots a'_n).$$

Then we consider the inverse of a_2 . It cannot be a'_1 , otherwise $a_2 = (a'_1)^{-1} = (a_1^{-1})^{-1} = a_1$, a contradiction. Without loss of generality we may call its inverse $a'_2 = a_2^{-1}$. Now

$$(a_2 \dots a_n)(a'_2 \dots a'_n) = (a_2 a'_2)(a_3 \dots a_n)(a'_3 \dots a'_n) = (a_3 \dots a_n)(a'_3 \dots a'_n).$$

Continue in this fashion we can cancel all terms and conclude that $x^2 = e$. $\square$

Proof. (simpler)

The inverse

$$\begin{aligned} i : G &\longrightarrow G \\ a &\longmapsto i(a) = a^{-1} \end{aligned}$$

is a bijection. Since G is abelian,

$$(a_1 a_2 \dots a_n) = (a_1^{-1} a_2^{-1} \dots a_n^{-1}).$$

Now

$$\begin{aligned} x^2 &= (a_1 a_2 \dots a_n)(a_1 a_2 \dots a_n) \\ &= (a_1 a_2 \dots a_n)(a_1^{-1} a_2^{-1} \dots a_n^{-1}) \\ &= (a_1 a_1^{-1}) \dots (a_n a_n^{-1}) \\ &= e. \end{aligned}$$

$\square$

4. 18. Using the results of Problems 15 and 16, prove that if p is an odd prime number, then $(p-1)! \equiv -1 \pmod{p}$. (This is known as Wilson's Theorem.) It is, of course, also true if $p = 2$.

Proof. Consider the group U_p . Since p is prime, the elements of the group are exactly the natural numbers from 1 up to $p-1$. Now

$$(p-1)! = 1 \cdot 2 \cdot \dots \cdot (p-1) = a_1 a_2 \dots a_n = x.$$

By Exercise 16, $x^2 = 1$. By Exercise 15, any member of U_p that is not 1 or $p-1$ does not satisfy $x^2 \equiv_p 1$. Note that:

$$x^2 \equiv_p 1 \Leftrightarrow x^2 = e \Leftrightarrow x = x^{-1}.$$

Therefore all members except 1 or $p-1$ are not their own inverse. Hence they must be pairs that are inverse of each other. Because U_p has even

order, and there are even number of elements excluding 1 and $p - 1$, the remaining elements can all pair up correctly. Because U_p is abelian, we may group the pairs together and cancel them out. Now

$$(p - 1)! = 1 \cdot (p - 1) \equiv_p -1.$$

□

5. 27. If $aH = bH$ forces $Ha = Hb$ in G , show that $aHa^{-1} = H$ for every $a \in G$.

Proof. Note that

$$\begin{aligned} aH &= bH \\ \Leftrightarrow a^{-1}b &\in H \\ \Leftrightarrow b^{-1}a &\in H. \end{aligned}$$

Using the assumption we also have

$$Ha = Hb \Leftrightarrow ab^{-1} \in H.$$

We have proved that for all $a, b \in G$,

$$b^{-1}a \in H \Rightarrow ab^{-1} \in H.$$

Therefore, for all $a, b \in G$,

$$ab \in H \Leftrightarrow ba \in H.$$

Fix any $a \in G$. We pick arbitrary $h \in H$ and let $b = ah$. Now $a^{-1}b = a^{-1}ah = h \in H$. Hence $ba^{-1} \in H$. Since $a(a^{-1}b) \in aH$, we have

$$\begin{aligned} a(a^{-1}b) &= (aa^{-1})b \\ &= b \\ &= b(a^{-1}a) \\ &= (ba^{-1})a \\ &\in Ha. \end{aligned}$$

Since h is picked arbitrarily, b can take all elements from aH . Since $b \in Ha$, we have shown that $aH \subset Ha$. Apply the same argument to the other side symmetrically, we obtain

$$aH = Ha$$

which means $aHa^{-1} = H$.

□

6. 31. If $o(a) = m$ and $a^s = e$, prove that $m \mid s$.

Proof. We first introduce

Lemma 1. For a cyclic group with order n and generating element a , we have for all integers q and r ,

$$a^q = a^r \Leftrightarrow q \equiv r \pmod{n}.$$

Proof of lemma.

$$\begin{aligned} a^q = a^r &\Leftrightarrow (a^q)(a^r)^{-1} = e \\ &\Leftrightarrow a^{q-r} = e. \end{aligned}$$

Now we apply euclidean's algorithm to $q-r$, and obtain x and y such that $0 \leq y < n$, and

$$q - r = nx + y.$$

Now

$$a^{q-r} = a^{nx+y} = (a^n)^x a^y = e^x a^y = a^y = e.$$

Since $0 \leq y < n$, by properties of cyclic group, $y = 0$. Hence

$$q - r = nx \equiv 0 \pmod{n}.$$

Now we prove the other direction, and assume $q \equiv r \pmod{n}$. Apply euclidean algorithm to both q and r . We obtain integers x_q, x_r, y_q, y_r such that $0 \leq y_q < n, 0 \leq y_r < n$ and

$$q = nx_q + y_q, r = nx_r + y_r.$$

We have $q \equiv y_q \pmod{n}$ and $r \equiv y_r \pmod{n}$. Hence $y_q \equiv y_r \pmod{n}$. Since both of them are in $\mathbb{Z}_n$, $y_q = y_r$. Let $y = y_q$. Now

$$\begin{aligned} a^{q-r} &= a^{(nx_q+y)-(nx_r+y)} \\ &= a^{nx_q-nx_r} \\ &= a^{n(x_q-x_r)} \\ &= (a^n)^{x_q-x_r} \\ &= e^{x_q-x_r} \\ &= e. \end{aligned}$$

□

By assumption we have $o(a) = |\langle a \rangle| = m$. By Theorem 2.4.5, $a^m = e$. Apply the last lemma. Since $a^s = e = a^m$,

$$s \equiv m \pmod{m}.$$

Hence

$$s \equiv 0 \pmod{m}.$$

Which means $m|s$.

□

7. 37. In a cyclic group of order n , show that for each positive integer m that divides n (including $m = 1$ and $m = n$) there are $\varphi(m)$ elements of order m .

Proof. An element $a^k \in \langle a \rangle = G$ has order m iff

(a) $(a^k)^m = a^{km} = e$. In other words, $n|km$, and

(b) m is the smallest positive integer such that km is a multiple of n . In other words, for any $p < m$, we have $n \nmid kp$.

Note that $1 \leq m \leq n$ (since m divides n), and $0 \leq k \leq n-1$ (since a^k are distinct elements of G).

We introduce a claim that is central to the proof, which we prove later:

Claim. The two conditions above are met, if and only if

$$(k, n) = \frac{n}{m}.$$

Note that

$$(k, n) = \frac{n}{m} \Leftrightarrow \left(\frac{k}{n/m}, \frac{n}{n/m} \right) = \left(\frac{k}{n/m}, m \right) = 1.$$

Hence $o(a^k) = m \Leftrightarrow \left(\frac{k}{n/m}, m \right) = 1$. Since $k < n$, $\frac{k}{n/m} < m$. There are precisely $\varphi(m)$ choices of $\frac{k}{n/m}$ such that $\left(\frac{k}{n/m}, m \right) = 1$ and $\frac{k}{n/m} < m$. Each choice of $\frac{k}{n/m}$ then uniquely corresponds to a choice of k . $\square$

Notation. For integer m and prime x , write $d_x(m)$ for the *multiplicity* of x in the unique prime factorization of m . For simplicity of notation, we may write $d_x(m, n)$ for $d_x((m, n))$.

Lemma 2. (Properties of Multiplicity) The following statements hold:

- (a) $d_x(n) \geq 0$.
- (b) $d_x(n) \geq a \Leftrightarrow x^a | n$.
- (c) $d_x(nm) = d_x(n) + d_x(m)$.
- (d) $d_x\left(\frac{n}{m}\right) = d_x(n) - d_x(m)$ if $m | n$.
- (e) $d_x(m, n) = \min(d_x(m), d_x(n))$.
- (f) If $m | n$, then for any prime x one has $d_x(m) \leq d_x(n)$. Equivalently, if $d_x(m) > d_x(n)$ then $m \nmid n$.
- (g) $m | n$ if and only if for all prime x such that $x | m$, $d_x(m) \leq d_x(n)$.

Proof. (a) Explicitly stated in the definition of multiplicity.

- (b) Apply unique factorization to n . Then there exists primes $p_1, p_2, \dots, p_k$ (all distinct), where k is a natural number, and positive integers $a_1, \dots, a_k$. We have

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}.$$

Without loss of generality, let $x = a_1$. If $d_x(n) \geq a$, then $a_1 \geq a$. Hence there exists a nonnegative integer c such that $a_1 = a + c$. Now

$$n = x^{a_1} p_2^{a_2} \dots p_k^{a_k} = x^{a+c} p_2^{a_2} \dots p_k^{a_k} = x^a x^c p_2^{a_2} \dots p_k^{a_k}$$

Therefore $x^a | n$.

Now assume $x^a | n$ and we want to show $d_x(n) \geq a$. Since $p_2^{a_2}, \dots, p_k^{a_k}$ are all prime factors other than x , we have

$$x^a \mid p_1^{a_1}.$$

Hence $x^{a_1} \geq x^a$. Since $x > 0$, we have $d_x(n) = a_1 \geq a$, as desired.

- (c) We apply unique factorization to both n and m . Then there exists primes $p_1, p_2, \dots, p_k$ (all distinct), primes $q_1, q_2, \dots, q_l$ (all distinct), where k, l are natural numbers, and positive integers $a_1, \dots, a_k, b_1, \dots, b_l$. We have

$$\begin{aligned} n &= p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} \\ m &= q_1^{b_1} q_2^{b_2} \dots q_l^{b_l}. \end{aligned}$$

Now choose some x out of the ps and qs . Without loss of generality we let $x = p_1$. We have two cases.

Case 1: $q_1, \dots, q_l \neq x$. Now

$$nm = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} q_1^{b_1} q_2^{b_2} \dots q_l^{b_l}.$$

$p_2, \dots, p_k, q_1, \dots, q_l$ are not equal to p_1 . Hence $d_x(nm) = a_1$. Since $d_x(m) = 0$, we have $d_x(nm) = d_x(n) + d_x(m)$.

Case 2: There exists some positive integer $i \leq l$ such that $q_i = x$. Without loss of generality, we assume $i = 1$. Now we use commutativity to obtain

$$\begin{aligned} nm &= p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} q_1^{b_1} q_2^{b_2} \dots q_l^{b_l} \\ &= p_1^{a_1} q_1^{b_1} p_2^{a_2} \dots p_k^{a_k} q_2^{b_2} \dots q_l^{b_l} \\ &= x^{a_1} x^{b_1} p_2^{a_2} \dots p_k^{a_k} q_2^{b_2} \dots q_l^{b_l} \\ &= x^{a_1+b_1} p_2^{a_2} \dots p_k^{a_k} q_2^{b_2} \dots q_l^{b_l}. \end{aligned}$$

Now $d_x(nm) = a_1 + b_1 = d_x(n) + d_x(m)$, as desired.

- (d) Apply $\frac{n}{m}$ and m to the last property. The assumption $m|n$ ensures that $\frac{n}{m}$ is an integer.
- (e) We have $(m, n) \mid m$ and $(m, n) \mid n$. Hence $d_x(m, n) \leq d_x(m)$ and $d_x(m, n) \leq d_x(n)$. Hence $d_x(m, n) \leq \min(d_x(m), d_x(n))$. Without loss of generality, let $\min(d_x(m), d_x(n)) = d_x(m)$. Suppose $d_x(m, n) < d_x(m)$, then there exists a positive integer c , such that $d_x(m, n) + c = d_x(m)$. Now

$$x^{d_x(m)} = x^{d_x(m, n) + c} \mid m.$$

$x^{d_x(m, n) + c} \mid n$ also because $d_x(n) \geq d_x(m)$. This implies $(m, n)x^c \mid m$ and $(m, n)x^c \mid n$, which is a contradiction since (m, n) is the greatest common divisor.

We have ruled out all other possibilities, so $d_x(m, n) = d_x(m) = \min(d_x(m), d_x(n))$.

- (f) Fix prime x . Use property (b) and taking $a = d_x(m)$, we have $x^{d_x(m)} | m$. By transitivity of divisibility, $x^{d_x(m)} | n$. Use property (b) again, we have $d_x(n) \geq d_x(m)$.
- (g) suppose $d_x(m) \leq d_x(n)$ for all primes x such that $x | m$. We write m in terms of its unique factorization:

$$m = q_1^{a_1} q_2^{a_2} \dots q_r^{a_r}$$

where $q_1, \dots, q_r$ are distinct primes, and $a_1, \dots, a_r$ are positive integers.

Now $a_1 = d_{q_1}(m) \leq d_{q_1}(n)$, hence $q_1^{a_1} | n$. Continue in this fashion, we see that $q_2^{a_2}, \dots, q_r^{a_r}$ all divide n . Therefore, $m | n$.

We now prove the other direction. Assume $m | n$, and let x be a prime such that $x | m$. Because $m | n$, there exists positive integer c such that $n = cm$. Hence

$$d_x(n) = d_x(cm) = d_x(c) + d_x(m) \geq d_x(m).$$

□

Lemma 3. If $p < m$, then there exists a prime $x > 1$ such that $x | m$ and $d_x(m) > d_x(p)$.

Proof. Fix p and m . Suppose for contradiction that for all $x | m$ and $x > 1$, we have $d_x(m) \leq d_x(p)$. Then apply property (f) above, we obtain that $m | p$. This means $p \geq m$, a contradiction to the assumption $p < m$. □

Proof of claim. Suppose $(k, n) = \frac{n}{m}$. Hence $(km, nm) = n$. This means that $n | km$.

Suppose there exists positive integer p such that $p < m$ and $n | kp$. Use the above lemma to choose prime x such that $x > 1$, $x | m$ and $d_x(m) > d_x(p)$. We claim that

$$d_x(n) = d_x(m) + d_x(k).$$

To verify this, since $(k, n) = \frac{n}{m}$, we have

$$\min(d_x(k), d_x(n)) = d_x(n) - d_x(m).$$

Do some arithmetic manipulations and obtain

$$d_x(m) = \max(0, d_x(n) - d_x(k)).$$

Since $x | m$, $d_x(m) \geq 1$, therefore

$$d_x(m) = d_x(n) - d_x(k).$$

Now

$$d_x(pk) = d_x(p) + d_x(k) < d_x(m) + d_x(k) = d_x(n).$$

Hence $n \nmid pk$.

For the other direction, assume that a^k has order m and hence satisfies the two conditions. We first introduce

Lemma 4. For any positive integer n, m, k , we have

$$n|mk \Rightarrow n|m(k, n).$$

Proof. Let x be any prime. Since $n|mk$,

$$d_x(n) \leq d_x(mk) = d_x(m) + d_x(k).$$

Now

$$\begin{aligned} d_x(m(k, n)) &= d_x(m) + d_x(k, n) \\ &= d_x(m) + \min(d_x(k), d_x(n)). \end{aligned}$$

When $\min(d_x(k), d_x(n)) = d_x(k)$, $d_x(m(k, n)) = d_x(m) + d_x(k) \geq d_x(n)$. When $\min(d_x(k), d_x(n)) = d_x(n)$, obviously we have $d_x(m(k, n)) = d_x(m) + d_x(n) \geq d_x(n)$. Hence in both cases, $d_x(m(k, n)) \geq d_x(n)$. Since this is true for all x , we have $n|m(k, n)$. $\square$

Then since $n|mk$, we apply the previous lemma and conclude that $n|m(k, n)$. Hence there exists positive integer c such that $cn = m(k, n)$. We claim that $c = 1$, which will conclude the proof.

To verify that $c = 1$, suppose for the sake of contradiction that $c > 1$. Now $(k, n) = c \frac{n}{m}$ since $m|n$. Hence c is a factor of both k and n , and we have

$$\left(\frac{k}{c}, \frac{n}{c}\right) = \frac{n}{m}.$$

Hence in particular

$$\frac{n}{m} \mid \frac{k}{c}.$$

Hence

$$n \mid m \frac{k}{c}$$

where $\frac{k}{c} < k$, a contradiction to the condition that k is the smallest integer such that $n|mk$. $\square$

Herstein, section 2.5, page 73 : Questions 2,6,7

8. 2. Recall that $G \simeq G'$ means that G is isomorphic to G' . Prove that for all groups G_1, G_2, G_3 :
 - (a) $G_1 \simeq G_1$.
 - (b) $G_1 \simeq G_2$ implies that $G_2 \simeq G_1$.
 - (c) $G_1 \simeq G_2, G_2 \simeq G_3$ implies that $G_1 \simeq G_3$.

Proof. (a) Take the identity mapping $\iota_G : G \rightarrow G$. Now ι is clearly bijective, and

$$\iota_G(a_1 a_2) = a_1 a_2 = \iota_G(a_1) \iota_G(a_2).$$

- (b) Since $G_1 \simeq G_2$, there exists bijective mapping $\varphi : G_1 \rightarrow G_2$ such that for all $a, b \in G_1$, $\varphi(ab) = \varphi(a)\varphi(b)$. Now consider the inverse mapping φ^{-1} . Clearly this mapping is bijective. For all elements

$x, y \in G_2$, let $a = \varphi^{-1}(x)$ and $b = \varphi^{-1}(y)$. Now $\varphi(a) = x$ and $\varphi(b) = y$. Now

$$\begin{aligned}\varphi^{-1}(xy) &= \varphi^{-1}(\varphi(a)\varphi(b)) \\ &= \varphi^{-1}(\varphi(ab)) \\ &= ab \\ &= \varphi^{-1}(\varphi(a))\varphi^{-1}(\varphi(b)) \\ &= \varphi^{-1}(x)\varphi^{-1}(y).\end{aligned}$$

- (c) Since $G_1 \simeq G_2$, there exists bijective mapping $\varphi : G_1 \rightarrow G_2$ such that for all $a, b \in G_1$, $\varphi(ab) = \varphi(a)\varphi(b)$. Since $G_2 \simeq G_3$, there exists bijective mapping $\psi : G_2 \rightarrow G_3$ such that for all $a, b \in G_2$, $\psi(ab) = \psi(a)\psi(b)$.

Now consider the composition $\psi \circ \varphi : G_1 \rightarrow G_3$. Clearly it is bijective since it's composition of two bijections. Now

$$\begin{aligned}\psi \circ \varphi(ab) &= \psi(\varphi(ab)) \\ &= \psi(\varphi(a)\varphi(b)) \\ &= \psi(\varphi(a))\psi(\varphi(b)) \\ &= \psi \circ \varphi(a)\psi \circ \varphi(b).\end{aligned}$$

□

9. 6. Prove that if $\varphi : G \rightarrow G'$ is a homomorphism, then $\varphi(G)$, the image of G , is a subgroup of G' .

Proof. (a) (Closure) Take $x, y \in \varphi(G)$. Then there exists $a, b \in G$ such that $\varphi(a) = x$ and $\varphi(b) = y$. Now

$$xy = \varphi(a)\varphi(b) = \varphi(ab) \in \varphi(G).$$

- (b) (Associativity) Inherited from G .

- (c) (Identity) Let $e \in G$ be the identity of G . Take $x \in \varphi(G)$. Now

$$\varphi(e)x = \varphi(e)\varphi(a) = \varphi(ea) = \varphi(a) = x.$$

Similarly, $x\varphi(e) = x$. Hence $\varphi(e)$ is an identity of $\varphi(G)$.

- (d) (Inverse) Take $x \in \varphi(G)$. Then $x = \varphi(a)$ for some $a \in G$. Now

$$\varphi(a)(\varphi(a^{-1})) = (\varphi(a^{-1}))\varphi(a) = \varphi(aa^{-1}) = \varphi(e).$$

Hence $\varphi(a^{-1})$ is a inverse for $\varphi(a)$.

□

Proof. (Simpler). Take $\varphi(a), \varphi(b) \in \varphi(G)$. We want to show $\varphi(a)\varphi(b)^{-1} \in \varphi(G)$. It suffices to show $\varphi(b)^{-1} = \varphi(b^{-1})$, then

$$\varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \in \varphi(G).$$

To show this, note that

$$\varphi(b)\varphi(b^{-1}) = \varphi(bb^{-1}) = \varphi(e).$$

Also, $\varphi(e)$ is the identity for $\varphi(G)$ since for any $c \in G$,

$$\varphi(e)\varphi(c) = \varphi(ec) = \varphi(c) = \varphi(c)\varphi(e).$$

□

10. 7. Show that $\varphi : G \rightarrow G'$, where φ is a homomorphism, is a monomorphism if and only if $\text{Ker } \varphi = (e)$.

Proof. Let e, e' be identity of G and G' respectively. If φ is a monomorphism, then the mapping is injective. For any element $a \in G$ such that $a \neq e$, we must have $\varphi(a) \neq \varphi(e) = e'$, hence $a \notin \text{Ker } \varphi$.

If φ is not a monomorphism, then there exists two distinct elements $a, b \in G$ such that $\varphi(a) = \varphi(b)$. Now $\varphi(ba^{-1}) = \varphi(b)\varphi(a^{-1}) = \varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e) = e'$. But $ba^{-1} \neq e$ since $a \neq b$. We have constructed a nonidentity element that is also in $\text{Ker } \varphi$. □